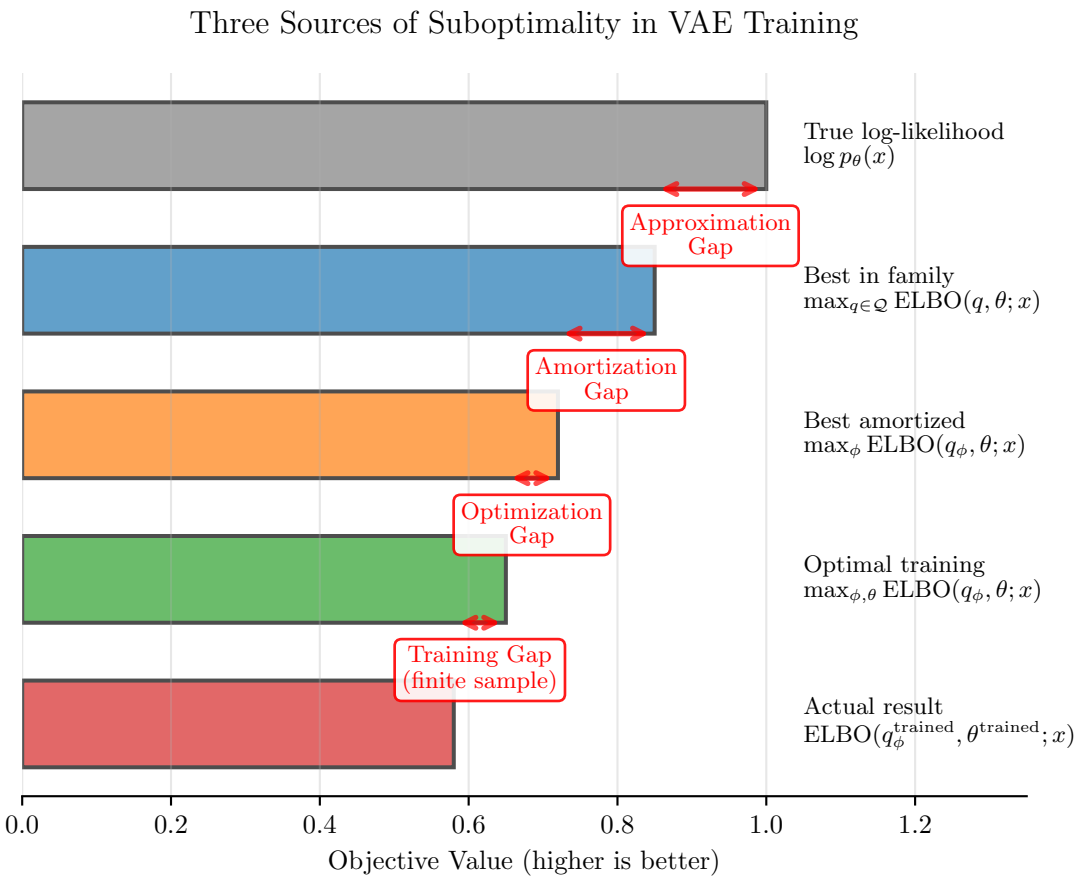
Three Sources of Suboptimality in VAE Training
True log-likelihood
$\log p_\theta(x)$
Approximation Gap
Best in family
$\max_{q \in \mathcal{Q}} \mathrm{ELBO}(q, \theta; x)$
Amortization Gap
Best amortized
$\max_\phi \mathrm{ELBO}(q_\phi, \theta; x)$
Optimization Gap
Optimal training
$\max_{\phi,\theta} \mathrm{ELBO}(q_\phi, \theta; x)$
Training Gap (finite sample)
Actual result
$\mathrm{ELBO}(q_\phi^{\text{trained}}, \theta^{\text{trained}}; x)$
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Objective Value (higher is better)